

Introduction to Microeconomics

Lecture 16—Production

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The production decisions of a firm

The production decisions of firms can be understood in three steps:

Production Technology: How inputs (such as labor, capital, and raw materials) can be transformed into outputs.

Cost Constraints: Given a target level of production, how costly it is to produce that level by choosing inputs.

Input Choices: Just as a consumer is constrained by a limited budget, the firm is concerned about the cost of production and the prices of labor, capital, and other inputs.

The **Theory of the Firm** is an explanation of how a firm makes cost-minimizing production decisions and how its cost varies with its output.

Factors of Production

Inputs into the production process (e.g., labor, capital, and materials). We can divide inputs into the broad categories of labor, materials and capital, each of which might include more narrow subdivisions. Labor inputs include skilled workers (carpenters, engineers) and unskilled workers (agricultural workers), as well as the entrepreneurial efforts of the firm's managers. Materials include steel, plastics, electricity, water, and any other goods that the firm buys and transforms into final products. Capital includes land, buildings, machinery and other equipment, as well as inventories.

The Technology of Production

Production function

The firm's **production function** for a particular good q

$$q = F(K, L),$$

shows the maximum amount of the good that can be produced using alternative combinations of K and L .

Production functions describe what is **technically feasible** when the firm operates **efficiently**.

Short Run vs Long Run

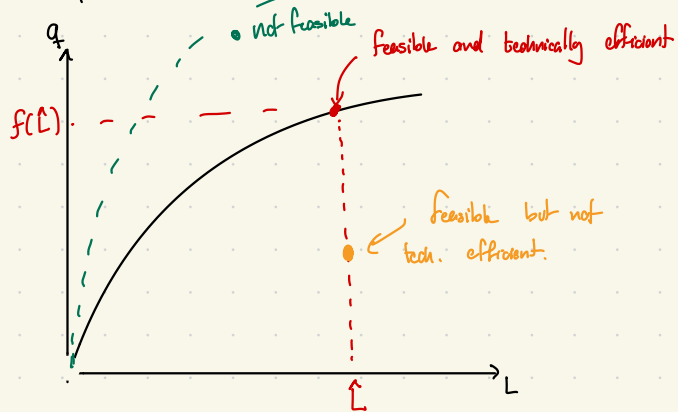
Short Run Period of time in which quantities of one or more production factors cannot be changed.

- A **fixed input** is a production factor that cannot be varied.

Long Run Period of time in which all inputs are variable.

$$q = f(L) = \sqrt{L}$$

← Technical progress



Marginal Physical Product

The **marginal physical product** of an input is the additional output that can be produced by using one more unit of that input while holding all other inputs constant.

For example, for a two input technology involving capital and labor $q = F(K, L)$:

$$MP_K = \frac{\Delta F(K, L)}{\Delta K}, \quad MP_L = \frac{\Delta F(K, L)}{\Delta L}$$

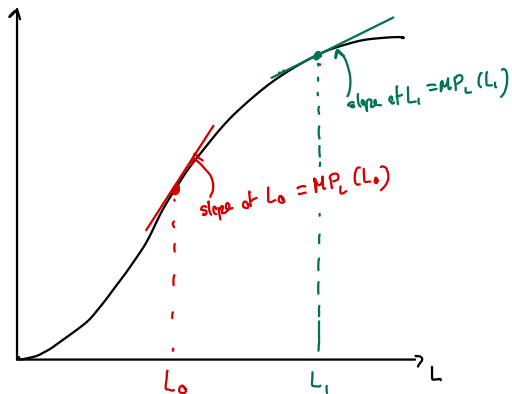
Diminishing Marginal Productivity?

- We might expect the marginal physical product of an input to depend on how much of that input is used.
- Example: labor cannot be added indefinitely to a given field.
- Formally, we expect that the rate of change of the MP with respect to changes in input to be decreasing.
- But changes on the marginal product do not only depend on how the input is used but also to **how other inputs are used**, i.e., they depend on the **input mix**, as well as on **technological growth**.
- Why has labor productivity, i.e., the MP_L risen over time?

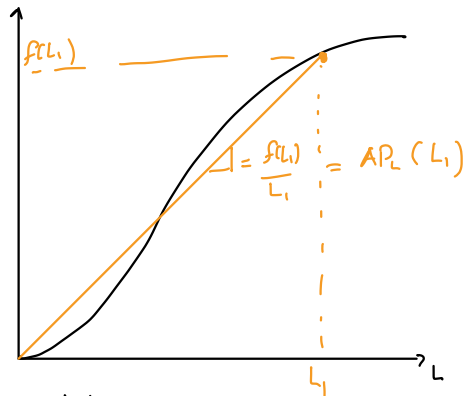
- In common usage, **productivity** often means **average productivity**.
- In layman parlance, increases in productivity are taken to be increases in **average productivity**.
- From the very first part of the course we know that in economic discussions and analysis the relevant concept is marginal, not average productivity.
- However, average productivity is easily measured.
- We define the average products to be:

$$AP_K = \frac{F(K, L)}{K} \quad AP_L = \frac{F(K, L)}{L}$$

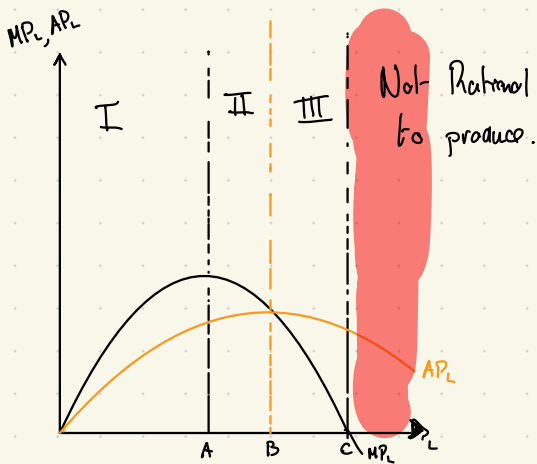
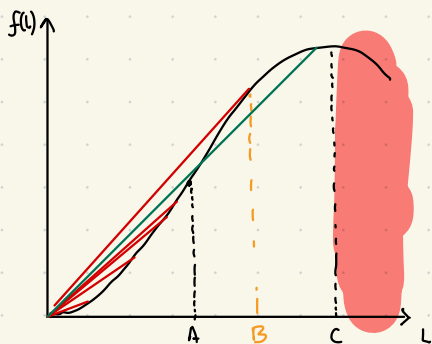
Relationship between average and marginal products



Graphical
computation
of MP_L M1



Graphical
derivation
of
 $AP_L = \frac{f(L)}{L} = \frac{f(L) - 0}{L - 0}$



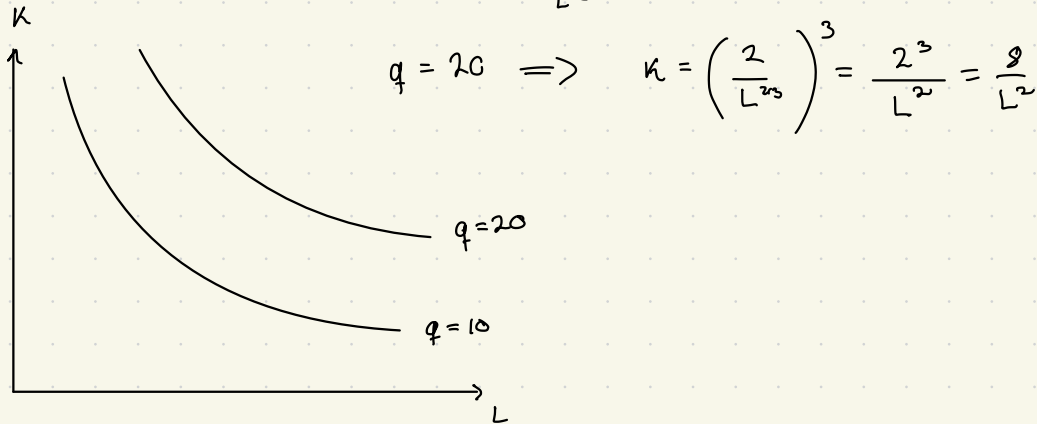
Isoquant

An **isoquant** shows those combinations of K and L that can produce a given level of output, say q_0 . Mathematically, is the set of input combinations that satisfy the equation:

$$q_0 = F(K, L)$$

Example: $q = 10 K^{1/3} L^{2/3}$ (Cobb-Douglas Production function)

$$q = 10 : \text{ Isoquant} \quad 10 = 10 K^{1/3} L^{2/3}$$
$$\Rightarrow K^{1/3} = \frac{1}{L^{2/3}} \Rightarrow K = \frac{1}{L^2}$$

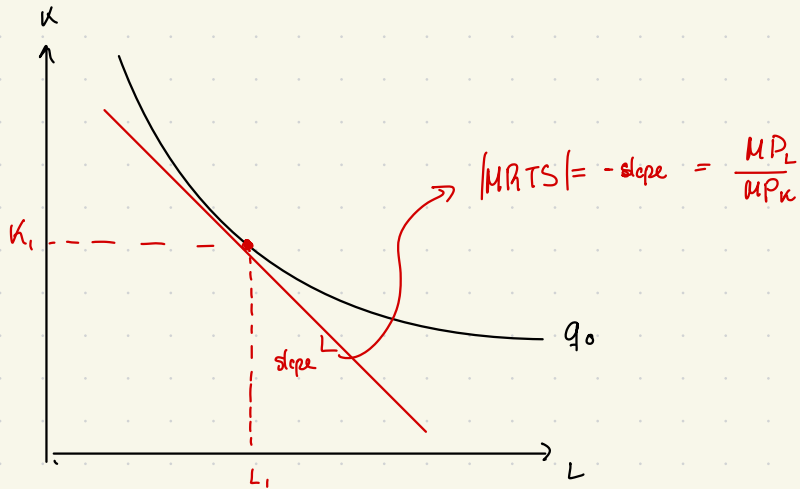


The marginal rate of technical substitution (MRTS)

Marginal rate of technical substitution

The **marginal rate of technical substitution (MRTS)** of K for L shows the rate at which, having added an extra unit of labor, capital can be decreased while holding output constant along an isoquant. Mathematically

$$MRTS = \frac{MP_L}{MP_K} = -\text{slope of the isoquant when } q = q_0$$



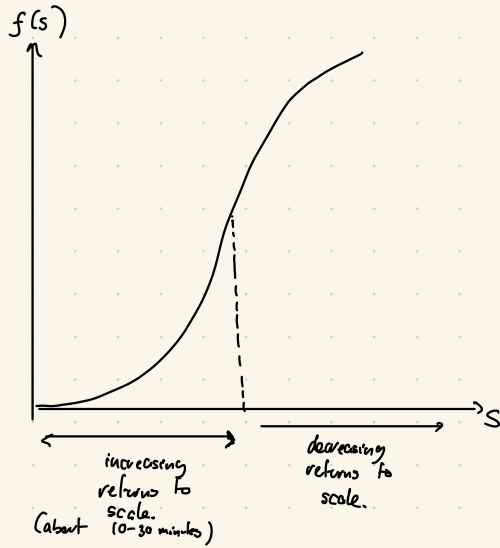
Returns to scale

If the production function given by $q = F(K, L)$ and if all inputs are multiplied by the same positive constant t , (where $t > 1$), then we classify **returns to scale** of the production function by

- **Constant returns to scale** $F(tK, tL) = tF(K, L) = tq$
- **Decreasing returns to scale** $F(tK, tL) < tF(K, L) = tq$
- **Increasing returns to scale** $F(tK, tL) > tF(K, L) = tq$

$f(s)$ = academic performance

s = study time in minutes



$$q = A K^{\alpha} L^{\beta} = F(K, L)$$

$\hookrightarrow A$: technology parameter

α : "determinant" capital

β : "det" of labor

$t > 1$: scaling factor

$$\begin{aligned} F(tK, tL) &= A (tK)^{\alpha} (tL)^{\beta} \\ &= A t^{\alpha} K^{\alpha} t^{\beta} L^{\beta} \\ &= t^{\alpha} t^{\beta} \underbrace{A K^{\alpha} L^{\beta}} \\ &= t^{\alpha+\beta} F(K, L) \end{aligned}$$

- If $\alpha + \beta = 1$ CRS
- If $\alpha + \beta < 1$ DRS
- If $\alpha + \beta > 1$ IRS.

$$F(K, L) = KL + K$$

$$\begin{aligned} F(tK, tL) &= (tK)(tL) + tK \\ &= t^2 KL + tK \end{aligned}$$

$$= tK (tL + 1) \quad \text{X}$$

Elasticity of Substitution

For the production function $q = F(K, L)$, the **elasticity of substitution** σ measures the proportionate change in K/L relative to the proportionate change in the MRTS along an isoquant. That is

$$\sigma = \frac{\Delta\%(K/L)}{\Delta\%MRTS}$$

$$\epsilon_{y,x} = \frac{\frac{\Delta \% y}{\Delta \% x}}{\frac{y_f - y_i}{x_f - x_i}} = \frac{\frac{y_f - y_i}{y_i}}{\frac{\Delta \ln(y)}{\Delta \ln(x)}}$$

Consider the arithmetic return: y_f, y_i

$$h := \frac{y_f - y_i}{y_i}$$

$$\Rightarrow y_f = y_i + h \cdot y_i = (1+h)y_i$$

Take $\ln$

$$\Rightarrow \ln y_f = \ln(1+h) + \ln(y_i)$$

$$\Rightarrow \ln(y_f) - \ln(y_i) = \ln(1+h)$$

$$\Delta \ln(y) = \ln(1+h) = h + \text{Error}$$

h is small $\approx h$
(close to zero)

When h is "small" $\ln(1+h)$ is approximated by h



Application: Calculation of the elasticity of substitution.

Usually, for the "standard" production functions we will work with we will have that

$$MRTS = f\left(\frac{K}{L}\right), \text{ where } f \text{ is a "simple" function}$$

"The MRTS is only a ^{simple} function of $\frac{K}{L}$, the capital/labor ratio."

Thus, to compute easily the elasticity of substitution, $\sigma = \frac{\Delta \ln(K/L)}{\Delta \ln(MRTS)}$, we just need to take the natural log of the equation above:

$$\ln(MRTS) = \ln(f(K/L))$$

when f is 'simple' the above can be simplified as

$$\ln(MRTS) = A + B \ln\left(\frac{K}{L}\right) \text{ where } A \text{ and } B \text{ are constants.}$$

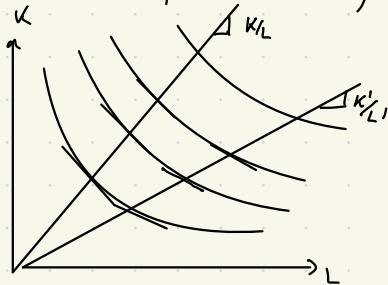
$$\text{Thus } \sigma = \frac{1}{B}$$

Case 1: $q = A K^\alpha L^\beta$

$$\left. \begin{aligned} MP_K &= A \alpha K^{\alpha-1} L^\beta \\ MP_L &= A \beta K^\alpha L^{\beta-1} \end{aligned} \right\} MRTS = \frac{MP_L}{MP_K} = \frac{\beta}{\alpha} \frac{K}{L}$$

$\hookrightarrow$ Take $\ln$ on both sides: $\ln(MRTS) = \ln\left(\frac{\beta}{\alpha}\right) + \ln\left(\frac{K}{L}\right)$

$$\frac{\Delta \ln(K/L)}{\Delta \ln(MRTS)} = 1 = \sigma \text{ (Elasticity of substitution)}$$



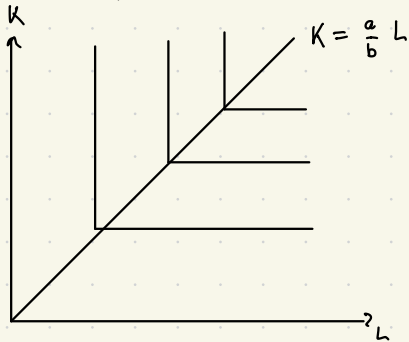
Case 2: $q = F(K, L) = aK + bL$, $a, b > 0$

$$\left. \begin{array}{l} MP_L = b \\ MP_K = a \end{array} \right\} MRTS = \frac{b}{a}$$

$$\ln(MRTS) = \ln\left(\frac{b}{a}\right) + 0 \times \ln\left(\frac{K}{L}\right)$$

$$\Rightarrow \sigma = \infty$$

Case 3: $q = \min \left\{ \frac{K}{a}, \frac{L}{b} \right\}$ $a, b > 0$



At the kink the MRTS is undefined.

$$\sigma = 0$$

Case 4: CES

constant elasticity substitution

$$q = (K^\delta + L^\delta)^{\frac{\sigma}{\delta}}$$

$$MP_L = \frac{\sigma}{\delta} (K^\delta + L^\delta)^{\frac{\sigma}{\delta} - 1} \delta L^{\delta-1}$$

$$MP_K = \frac{\sigma}{\delta} (K^\delta + L^\delta)^{\frac{\sigma}{\delta} - 1} \delta K^{\delta-1}$$

$$MRTS = \frac{MP_L}{MP_K} = \frac{L^{\delta-1}}{K^{\delta-1}} = \left(\frac{K}{L}\right)^{1-\delta}$$

$$\ln(MRTS) = (1-\delta) \ln\left(\frac{K}{L}\right) \Rightarrow \ln\left(\frac{K}{L}\right) = \frac{1}{1-\delta} \ln(MRTS)$$

$\sigma = \frac{1}{1-\delta}$, thus, the elasticity of substitution is constant

CES encapsulates:

- Cobb-Douglas
- Perfect Complements
- Perfect Substitutes

δ = parameter associated with the elasticity of substitution

σ = parameter associated with returns to scale.